Crypto Shocks

Ambrogio Cesa-Bianchi
Bank of England, CEPR, and CfM

Marco Pinchetti
Banque de France

ESSIM
29 May 2025

*The views expressed in this paper are those of the author(s) and do not necessarily represent the views of the BoE, BdF, or the Eurosystem.

What is this paper about?

What is this paper about?

- Whatever you're thinking about, not this...

What is this paper about?

- **Question** Do crypto-currency fluctuations propagate to the rest of the economy?

What is this paper about?

- **Question** Do crypto-currency fluctuations propagate to the rest of the economy?
- Well documented correlation between Bitcoin and S&P500 returns [Adrian et al. (2022)]
- Two potential directions of causality:
  - Macro and policy shocks → Crypto-currency prices
  - Crypto-currency shocks → Macro/financial fluctuations

What is this paper about?

- **Question** Do crypto-currency fluctuations propagate to the rest of the economy?
- Well documented correlation between Bitcoin and S&P500 returns [Adrian et al. (2022)]
- Two potential directions of causality:
  - Macro and policy shocks → Crypto-currency prices
  - Crypto-currency shocks → Macro/financial fluctuations
- **Policy relevance** Crypto-to-macro effects potentially relevant for
  - Regulatory and supervisory considerations
  - Conduct of monetary policy

What we do, what we find

- **Approach**
  [1] Construct a daily series of exogenous crypto surprises
  [2] Develop an ‘agnostic’ indicator of exposure to crypto currencies at firm-level
  [3] Evaluate the effects of crypto surprises on firm-level equity prices
  [4] Disentangle transmission mechanisms

What we do, what we find

- **Approach**
  [1] Construct a daily series of exogenous crypto surprises
  [2] Develop an 'agnostic' indicator of exposure to crypto currencies at firm-level
  [3] Evaluate the effects of crypto surprises on firm-level equity prices
  [4] Disentangle transmission mechanisms

- **Results**
  * Crypto surprises affect crypto-exposed stocks (the more so the higher their exposure)
  * Multiple mechanisms at work

What we do, what we find

- **Approach**
  [1] Construct a daily series of exogenous crypto surprises
  [2] Develop an 'agnostic' indicator of exposure to crypto currencies at firm-level
  [3] Evaluate the effects of crypto surprises on firm-level equity prices
  [4] Disentangle transmission mechanisms

- **Results**
  * Crypto surprises affect crypto-exposed stocks (the more so the higher their exposure)
  * Multiple mechanisms at work (balance sheet effects most powerful)

Preliminary!

Literature

- **Narrative identification**
  Romer and Romer (2004), Romer and Romer (2010), Cloyne (2013), Cloyne and Hurtgen (2016), many others

- **High frequency identification & Event Studies**
  Gürkaynak, Sack, and Swanson (2005), Gurkaynak, Kisacikoglu, and Wright (2020), Känzig (2021), many others

- **Crypto-currencies and the macroeconomy**
  - Macro to Crypto
    Karau (2023), Che, Copestake, Furceri, and Terracciano (2023), Divakaruni and Zimmerman (2024), Benigno and Rosa (2023)
  - Crypto to Macro
    Aiello, Baker, Balyuk, DiMaggio, Johnson, and Kotter (2023), Weber, Candia, Coibion, and Gorodnichenko (2023)

Identifying Crypto Shocks

Combining narrative and high-frequency identification approaches

- **Narrative events**
  - Identify events that originate in the crypto-sphere...
  - ...but are unrelated to macro developments

Combining narrative and high-frequency identification approaches

- **Narrative events**
  - Identify events that originate in the crypto-sphere...
  - ...but are unrelated to macro developments
- **High-frequency surprises**
  - Compute Bitcoin spot price change in a short window around each event

Combining narrative and high-frequency identification approaches

- **Narrative events**
  - Identify events that originate in the crypto-sphere...
  - ...but are unrelated to macro developments
- **High-frequency surprises**
  - Compute Bitcoin spot price change in a short window around each event
- Resulting series of surprises (s_t) can be used to estimate the causal effects of crypto shocks

Step 1: Narrative approach

- Search for Bitcoin-related articles in Google News
  - Week by week from 2017
  - Based on selected keywords: Bitcoin, crypto, etc. See n-grams

Step 1: Narrative approach

- Search for Bitcoin-related articles in Google News
  - Week by week from 2017
  - Based on selected keywords: Bitcoin, crypto, etc. See n-grams
- Keep events that originated in crypto markets and had some effect on crypto prices
  - Narrative judgement

Step 1: Narrative approach

- Search for Bitcoin-related articles in Google News
  - Week by week from 2017
  - Based on selected keywords: Bitcoin, crypto, etc. See n-grams
- Keep events that originated in crypto markets and had some effect on crypto prices
  - Narrative judgement
- Save time stamp of the event (day t at time k)
  - Typically mentioned in the news article
  - Otherwise use the timestamp of the news article

Step 1: Narrative approach

- Search for Bitcoin-related articles in Google News
  - Week by week from 2017
  - Based on selected keywords: Bitcoin, crypto, etc. See n-grams
- Keep events that originated in crypto markets and had some effect on crypto prices
  - Narrative judgement
- Save time stamp of the event (day t at time k)
  - Typically mentioned in the news article
  - Otherwise use the timestamp of the news article
- **Final dataset** 151 crypto events between January 2017 and December 2022

Step 2: High-frequency approach

- **Data** Bitcoin prices sampled at 5-minute frequency
  - Source: Kraken (US–based cryptocurrency exchange)

Step 2: High-frequency approach

- **Data** Bitcoin prices sampled at 5-minute frequency
  - Source: Kraken (US–based cryptocurrency exchange)

- **Surprise** Price change using a 2-hour window around an event on day t at time k:

$$s_t = BTC_{t,k+60'} - BTC_{t,k-60'}$$

Step 2: High-frequency approach

- **Data** Bitcoin prices sampled at 5-minute frequency
  - Source: Kraken (US–based cryptocurrency exchange)

- **Surprise** Price change using a 2-hour window around an event on day t at time k:

$$s_t = BTC_{t,k+60'} - BTC_{t,k-60'}$$

- **Key assumption** No other shock systematically affects Bitcoin price within the window

$$\mathbb{E}[s_t, \epsilon_t^{Macro}] = 0$$

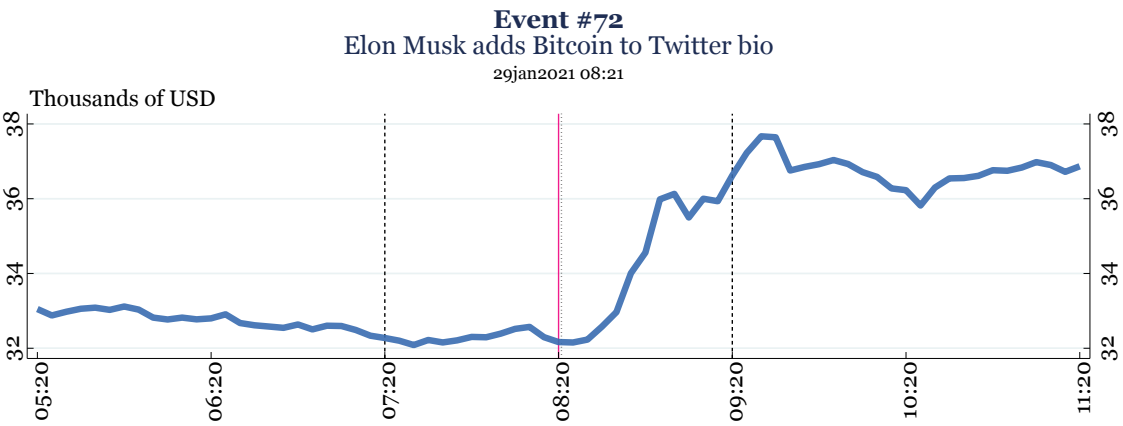
An example: Elon Musk endorses Bitcoin via his Twitter profile

- Date stamp: Friday 29 January 2022, at 8:21 AM (GMT)
- Elon Musk changes his Twitter profile to:

- Musk also tweets a cryptic message:

An example: Elon Musk endorses Bitcoin via his Twitter profile

Date stamp: 29th January 2021 at 8:21 AM (GMT)



Another example: Google bans crypto-currency adverts

- Date stamp: 14 March 2018 at around 4:00 AM (GMT)
- Excerpt from the article:

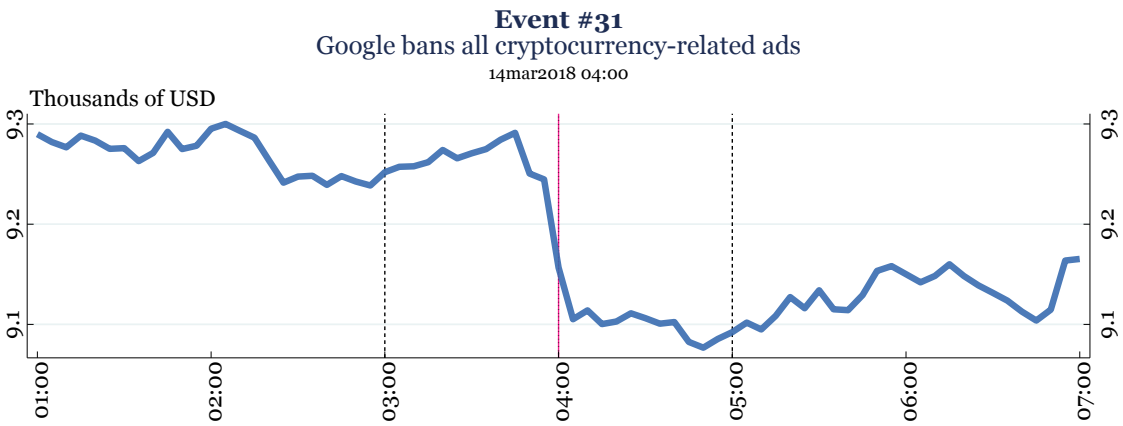
  "Google has moved to ban crypto-currency advertising from its platforms.

  [...]

  The value of Bitcoin, Ethereum and Ripple all dipped after the news emerged at about 04:00 GMT."

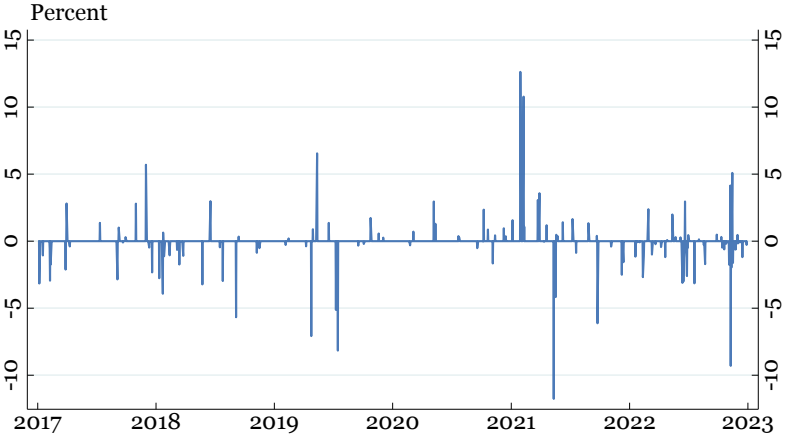
An example: Google bans crypto-currency adverts

Date stamp: 14th of March at 04:00 AM (GMT)



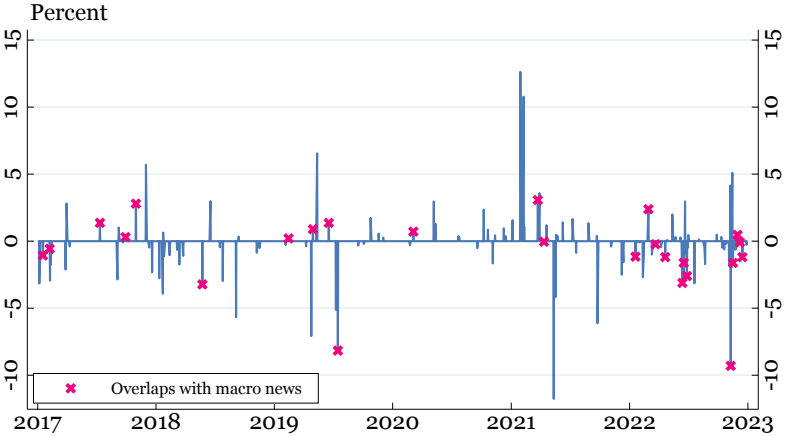
From events to a daily series of surprises

- Aggregate events to daily surprises, drop non business days (from 151 events to 133 surprises)



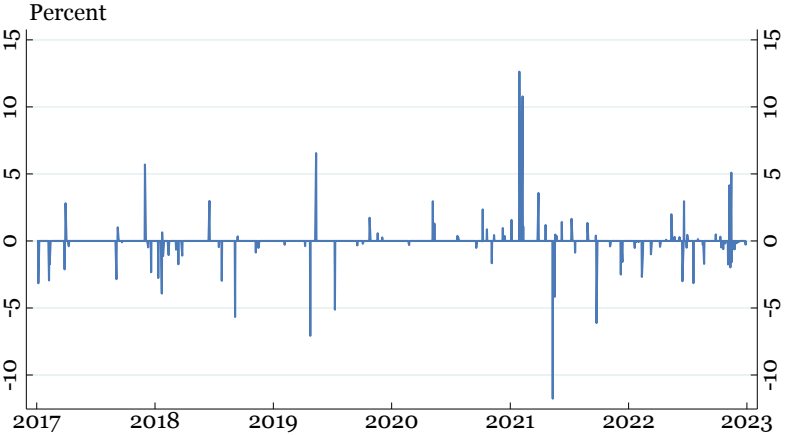
From events to a daily series of surprises

- Drop events overlapping with macro news from Bloomberg calendar (from 133 to 107 surprises)



The crypto surprise series

- Baseline series of surprises



Stats | Distribution | By Hour | By Quarter | By Year | Autocorrelation

[Sanity check] The effect of a crypto surprise on Bitcoin prices

- Estimate the following event-study specification:

$$\Delta BTC_t = \alpha + \beta s_t + \varepsilon_t$$

  - ΔBTC_t: daily Bitcoin price change
  - s_t: daily crypto surprise

[Sanity check] The effect of a crypto surprise on Bitcoin prices

- Estimate the following event-study specification:

$$\Delta BTC_t = \alpha + \beta s_t + \varepsilon_t$$

  - ΔBTC_t: daily Bitcoin price change
  - s_t: daily crypto surprise

	(1) ΔBTC_t
Crypto surprise (± 1-hour window)	0.97*** (0.24)
Observations	107
R squared	0.274
F-Statistic	15.7

NOTE. Robust standard errors are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Coefficients corresponding to the constant not reported.

[Sanity check] The effect of a crypto surprise on Bitcoin prices

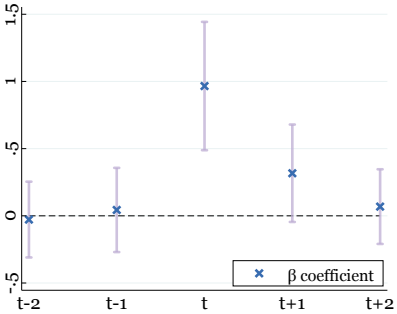
- Estimate the following event-study specification:

$$\Delta BTC_t = \alpha + \beta s_t + \varepsilon_t$$

  - ΔBTC_t: daily Bitcoin price change
  - s_t: daily crypto surprise

	(1) ΔBTC_t
Crypto surprise (± 1-hour window)	0.97*** (0.24)
Observations	107
R squared	0.274
F-Statistic	15.7

NOTE. Robust standard errors are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Coefficients corresponding to the constant not reported.



Firm Exposure to Crypto

Firm exposure to crypto

- **Question** Which firms are exposed to crypto fluctuations, and by how much?
- **Agnostic approach** Text analysis of listed firms' earnings calls via NL Analytics

Firm exposure to crypto

- **Question** Which firms are exposed to crypto fluctuations, and by how much?
- **Agnostic approach** Text analysis of listed firms' earnings calls via NL Analytics
- What are earnings calls?
  - Quarterly discussions between management and analysts
  - Held after public earnings releases
  - Focus on outlook, risks, and strategy (in addition to current earnings)

Details

Measuring crypto exposure

- Define set of n-grams associated with crypto discussions ($\mathbb{C}$) See crypto n-grams

Measuring crypto exposure

- Define set of n-grams associated with crypto discussions ($\mathbb{C}$) See crypto n-grams

- Let n_{cit} denote the sentences contained in the call c of firm i on day t:

$$\text{CryptoExpo}_{cit} = \frac{1}{N_{cit}} \sum_{n=1}^{N_{cit}} \mathbb{1}\left[n_{cit} \in \mathbb{C}\right]$$

Measuring crypto exposure

- Define set of n-grams associated with crypto discussions ($\mathbb{C}$) See crypto n-grams

- Let n_{cit} denote the sentences contained in the call c of firm i on day t:

$$\text{CryptoExpo}_{cit} = \frac{1}{N_{cit}} \sum_{n=1}^{N_{cit}} \mathbb{1}\left[n_{cit} \in \mathbb{C}\right]$$

- Let $C_{i\tau}$ be the number of earnings calls for firm i in days t belonging to quarter τ:

$$\text{CryptoExpo}_{i\tau} = \frac{1}{C_{i\tau}} \sum_{c=1}^{C_{i\tau}} \text{CryptoExpo}_{cit}$$

Measuring crypto exposure

- Define set of n-grams associated with crypto discussions ($\mathbb{C}$) See crypto n-grams
- Let n_{cit} denote the sentences contained in the call c of firm i on day t:

$$\text{CryptoExpo}_{cit} = \frac{1}{N_{cit}} \sum_{n=1}^{N_{cit}} \mathbb{1}\left[n_{cit} \in \mathbb{C}\right]$$

- Let $C_{i\tau}$ be the number of earnings calls for firm i in days t belonging to quarter τ:

$$\text{CryptoExpo}_{i\tau} = \frac{1}{C_{i\tau}} \sum_{c=1}^{C_{i\tau}} \text{CryptoExpo}_{cit}$$

- Let T_i be the number of quarters with earnings calls for firm i :

$$\text{CryptoExpo}_{i} = \frac{1}{T_i} \sum_{\tau=1}^{T_i} \text{CryptoExpo}_{i\tau}$$

Other measures: Risk & Sentiment

Crypto exposure: Sample selection & Details

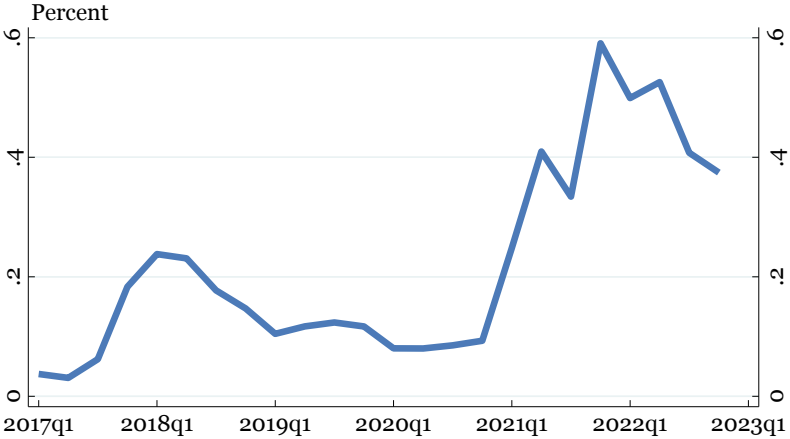
- Sample period from 1 Jan 2017 to 31 Dec 2022 (same as crypto surprises)
- Keep only firms with non-zero average exposure ($\text{CryptoExpo}_i \neq 0$)
- Baseline sample
  - Number of firms: 1, 221
  - Number of countries: 53

Crypto exposure: Sample selection & Details

- Sample period from 1 Jan 2017 to 31 Dec 2022 (same as crypto surprises)
- Keep only firms with non-zero average exposure ($\text{CryptoExpo}_i \neq 0$)
- Baseline sample
  - Number of firms: 1, 221
  - Number of countries: 53
- Robustness using sentiment measure ($\text{CryptoSent}_i \neq 0$)
  - Number of firms: 800
  - Number of countries: 53

Crypto exposure measure

Average and median across firms

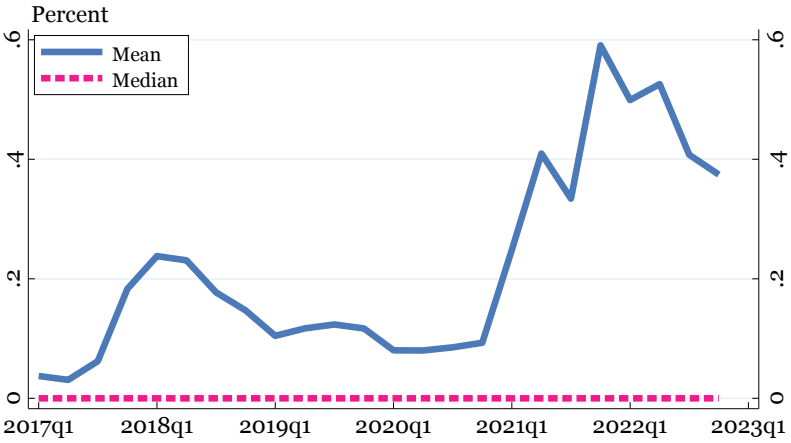


NOTE. Cross-sectional average (with equal weights) of the crypto-currency exposure measure $CryptoExpo_{it}$ across all firms in our sample. Sample period: 2017q1 to 2022q4.

Compare to sentiment

Crypto exposure measure

Average and median across firms

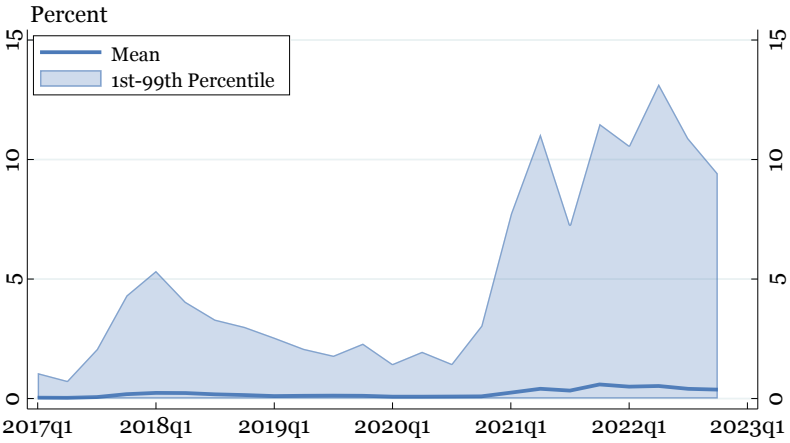


NOTE. Cross-sectional average (with equal weights) and median of the crypto-currency exposure measure $CryptoExpo_{it}$ across all firms in our sample. Sample period: 2017q1 to 2022q4.

Compare to sentiment

Crypto exposure measure

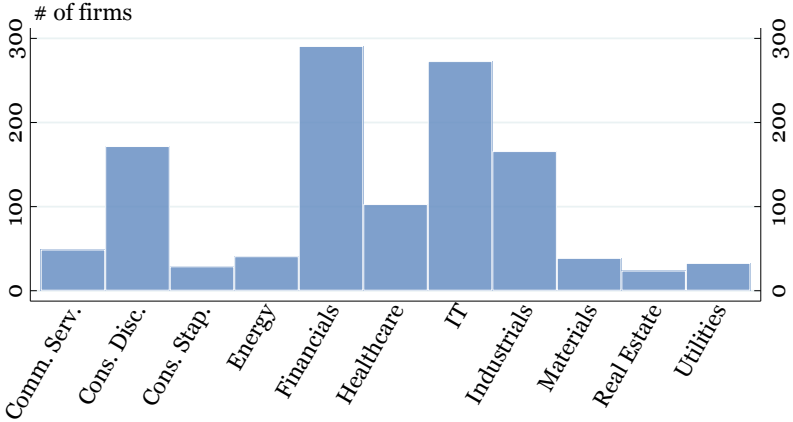
Cross-sectional distribution



NOTE. Cross-sectional average of the crypto-currency exposure measure $CryptoExpo_{it}$ (with equal weights) across all firms in our sample. Shaded areas are 1st and 99th percentiles. Sample period: 2017q1 to 2022q4.

Crypto exposure measure

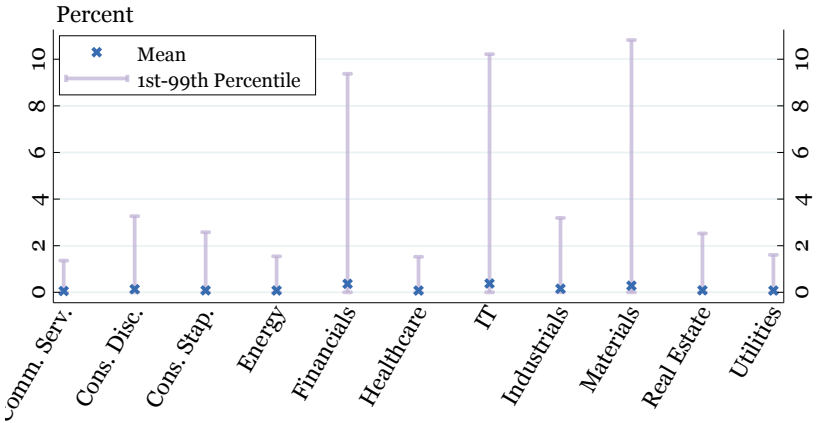
Number of firms by sector



NOTE. Number of firms with non-zero exposure by sector. Sample period: 2017q1 to 2022q4.

Crypto exposure measure

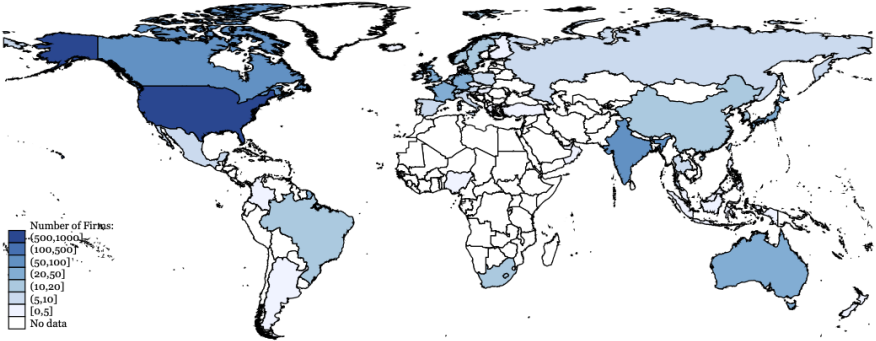
Sectoral breakdown: Exposure distribution



NOTE. Cross-sectional average of the crypto-currency exposure measure $CryptoExpo_{it}$ (with equal weights) across firms by sector. Whiskers are 1st and 99th percentiles. Sample period: 2017q1 to 2022q4.

Crypto exposure measure

Geographic breakdown: Number of firms



Exposure

Transmission

Impact Effects

Event study regressions

- **Objective** Estimate the high frequency effects of crypto surprises on equity prices
- **Event study data set** Time dimension t indexes days with crypto events

Event study regressions

- **Objective** Estimate the high frequency effects of crypto surprises on equity prices
- **Event study data set** Time dimension t indexes days with crypto events
- **Event study regressions**
  - Main focus: Equity price of crypto-exposed firms (average and heterogeneous effects)
  - Bonus: Aggregate equity indices & other asset prices

The effect of crypto surprises on firm-level equity prices

- Estimate variants of the panel event study regression:

$$y_{i,t+1} = \alpha_i + \beta s_t + \gamma \left(s_t \times CryptoExpo_{i,\tau-1}^{High} \right) + \delta CryptoExpo_{i,\tau-1}^{High} + \varepsilon_{i,t+1}$$

  * y_{it} 2-day log-change in firm i's equity price
  * α_i firm fixed effect
  * s_t daily crypto surprise
  * $CryptoExpo_{i\tau}^{High} = 1$ if $CryptoExpo_{i\tau}$ is above the median

The effect of crypto surprises on firm-level equity prices

- Estimate variants of the panel event study regression:

$$y_{i,t+1} = \alpha_i + \beta s_t + \gamma \left(s_t \times CryptoExpo_{i,\tau-1}^{High} \right) + \delta CryptoExpo_{i,\tau-1}^{High} + \varepsilon_{i,t+1}$$

  - y_{it} 2-day log-change in firm i's equity price
  - α_i firm fixed effect
  - s_t daily crypto surprise
  - $CryptoExpo_{i\tau}^{High} = 1$ if $CryptoExpo_{i\tau}$ is above the median

- $\boxed{\beta}$ captures average effect of crypto surprises on equity prices in the cross-section of firms

The effect of crypto surprises on firm-level equity prices

- Estimate variants of the panel event study regression:

$$y_{i,t+1} = \alpha_i + \beta s_t + \gamma \left(s_t \times CryptoExpo_{i,\tau-1}^{High} \right) + \delta CryptoExpo_{i,\tau-1}^{High} + \varepsilon_{i,t+1}$$

  - $y_{i,t+1}$ 2-day log-change in firm i's equity price
  - α_i firm fixed effect
  - s_t daily crypto surprise
  - $CryptoExpo_{i\tau}^{High} = 1$ if $CryptoExpo_{i\tau}$ is above the median

- $\boxed{\gamma}$ captures relative effect on the equity price of highly exposed firms

The effect of crypto surprises on firm-level equity prices

	(1)	(2)	(3)	(4)
Crypto surprise (β)	0.10**	0.10**		
	(0.05)	(0.05)		
Crypto surprise x High Exposure (γ)		0.07*	0.05**	0.06**
		(0.04)	(0.02)	(0.03)
High Exposure (δ)		-0.15	-0.05	-0.07
		(0.15)	(0.06)	(0.06)
Observations	113,006	108,085	108,085	98,458
R squared	0.02	0.02	0.20	0.32
Firm FE	yes	yes	yes	yes
Time x Sector FE	no	no	yes	no
Time x Sector x Country FE	no	no	no	yes

NOTE. Robust standard errors (clustered two-way, by firm and day) are reported in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Coefficients corresponding to the constant and fixed effects not reported.

The effect of crypto surprises on firm-level equity prices

Robustness: Average effects

	(1) Baseline	(2) With Overlap	(3) Pre 2020	(4) Post 2020	(5) Drop top 10 shocks	(6) IV
Crypto surprise (β)	0.10**	0.08	-0.08	0.16***	0.19*	
	(0.05)	(0.06)	(0.05)	(0.06)	(0.11)	
Bitcoin price log-change (β)						0.11*
						(0.05)
Observations	113,006	140,735	37,358	75,645	102,354	113,006
R squared	0.02	0.02	0.04	0.03	0.02	0.00
Firm FE	yes	yes	yes	yes	yes	yes
Time x Sector FE	no	no	no	no	no	no
Time x Sector x Country FE	no	no	no	no	no	no

NOTE. Robust standard errors (clustered two-way, by firm and day) are reported in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Coefficients corresponding to the constant and fixed effects not reported.

The effect of crypto surprises on firm-level equity prices

Robustness: Heterogeneity

	(1) Baseline	(2) With Overlap	(3) Pre 2020	(4) Post 2020	(5) Drop top 10 shocks	(6) IV
Crypto surprise x High Exposure (γ)	0.06**	0.06**	0.01	0.07**	0.05*	
	(0.03)	(0.02)	(0.04)	(0.03)	(0.03)	
Bitcoin price x High Exposure (γ)						0.07*
						(0.04)
Observations	98,458	122,035	29,475	68,981	97,447	98,458
R squared	0.32	0.35	0.28	0.34	0.32	-0.00
Firm FE	yes	yes	yes	yes	yes	yes
Time x Sector FE	no	no	no	no	no	no
Time x Sector x Country FE	yes	yes	yes	yes	yes	yes

NOTE. Robust standard errors (clustered two-way, by firm and day) are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Coefficients corresponding to the constant, fixed effects, and the dummy based on $CryptoExpo_{i\tau}$ not reported.

More on identification: Aggregate asset prices

- **Identification concern** Are we just capturing a market-wide factor?
  - If yes, crypto surprises should move broad asset prices

More on identification: Aggregate asset prices

- **Identification concern** Are we just capturing a market-wide factor?
  - If yes, crypto surprises should move broad asset prices
- Estimate the following event study regression:

$$y_{t+1} = \alpha + \beta s_t + \varepsilon_{t+1}$$

  - y_{t+1} 2-day (log) change in an aggregate asset price (e.g. S&P500, Vix Index, gold price, etc)
  - α constant
  - s_t daily crypto surprise

More on identification: Aggregate asset prices

- **Identification concern** Are we just capturing a market-wide factor?
  - If yes, crypto surprises should move broad asset prices
- Estimate the following event study regression:

$$y_{t+1} = \alpha + \beta s_t + \varepsilon_{t+1}$$

  - y_{t+1} 2-day (log) change in an aggregate asset price (e.g. S&P500, Vix Index, gold price, etc)
  - α constant
  - s_t daily crypto surprise
- $\boxed{\beta}$ captures effect of crypto surprises on asset price y_t

The effect of crypto surprises on aggregate asset prices

	(1) Crypto-firms Index	(2) S&P500 Index	(3) Vix Index	(4) 10-year Treasury Yield	(5) Dollar Exch. Rate	(6) Gold Price	(7) Corporate Bond Spread
Crypto surprise (β)	0.10**	0.07	0.03	-0.46	0.01	0.02	0.05
	(0.04)	(0.04)	(0.07)	(0.31)	(0.02)	(0.04)	(0.09)
Observations	107	107	107	107	107	107	107
R squared	0.02	0.01	0.00	0.02	0.00	0.00	0.00

NOTE. Robust standard errors are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Coefficients corresponding to the constant and fixed effects not reported.

The effect of crypto surprises on aggregate asset prices

	(1) Crypto-firms Index	(2) S&P500 Index	(3) Vix Index	(4) 10-year Treasury Yield	(5) Dollar Exch. Rate	(6) Gold Price	(7) Corporate Bond Spread
Crypto surprise (β)	0.10**	0.07	0.03	-0.46	0.01	0.02	0.05
	(0.04)	(0.04)	(0.07)	(0.31)	(0.02)	(0.04)	(0.09)
Observations	107	107	107	107	107	107	107
R squared	0.02	0.01	0.00	0.02	0.00	0.00	0.00

NOTE. Robust standard errors are reported in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Coefficients corresponding to the constant and fixed effects not reported.

The effect of crypto surprises on aggregate asset prices

Post 2020 sample

	(1) Crypto-firms Index	(2) S&P500 Index	(3) Vix Index	(4) 10-year Treasury Yield	(5) Dollar Exch. Rate	(6) Gold Price	(7) Corporate Bond Spread
Crypto surprise (β)	0.17***	0.13**	-0.03	-0.34	-0.01	0.04	-0.03
	(0.05)	(0.06)	(0.09)	(0.41)	(0.03)	(0.05)	(0.12)
Observations	68	68	68	68	68	68	68
R squared	0.05	0.03	0.00	0.01	0.00	0.01	0.00

NOTE. Robust standard errors are reported in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Coefficients corresponding to the constant and fixed effects not reported.

[Sanity check] The effect of S&P 500 on aggregate asset prices

- Regress same set of variables on S&P 500 returns (≈ market factor):

$$y_t = \alpha + \beta \Delta SP500_t + \varepsilon_t$$

	(1) Crypto-firms Index	(2) S&P500 Index	(3) Vix Index	(4) 10-year Treasury Yield	(5) Dollar Exch. Rate	(6) Gold Price	(7) Corporate Bond Spread
S&P500 return	0.94***	1.00***	-0.99***	0.44**	-0.13***	0.13***	-1.78***
	(0.03)	(0.00)	(0.09)	(0.18)	(0.01)	(0.04)	(0.42)
Observations	1,536	1,536	1,502	1,536	1,536	1,536	1,536
R squared	0.76	1.00	0.30	0.01	0.13	0.03	0.03

NOTE. Robust standard errors are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Coefficients corresponding to the constant and fixed effects not reported.

Transmission

Dynamics

Local projections

- **Objective** Estimate the propagation of crypto surprises
- **Weekly data set** Collapse data at weekly frequency
  - Sum of surprises by week Weekly surprise
  - Average of daily equity prices

Local projections

- **Objective** Estimate the propagation of crypto surprises
- **Weekly data set** Collapse data at weekly frequency
  - Sum of surprises by week Weekly surprise
  - Average of daily equity prices
- **Panel local projections**
  - Equity price of crypto-exposed firms, both average and heterogeneous effects

Panel local projections

- Estimate the following local projection specification:

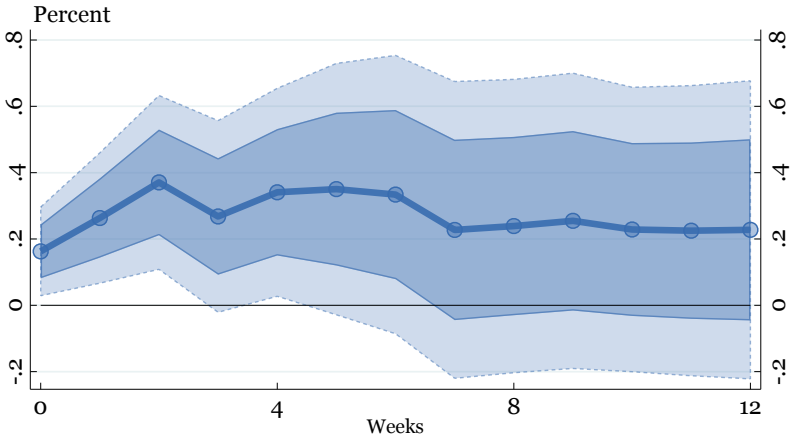
$$y_{it} = \alpha_i + \beta s_t + \sum_{p=0}^{4} \phi_p y_{i,t-p} + \sum_{p=0}^{4} \Gamma_p Z_{t-p} + \varepsilon_{it}$$

  - y_{it} weekly equity price for firm i
  - α_i firm fixed effect
  - s_t weekly crypto surprise
  - Z_t vector of controls (global economic surprise index, Vix index, BTC price, gold price, time trend)
  - Standard errors clustered two-way (firm and week)

- $\boxed{\beta}$ captures the average effect of crypto surprises on equity prices in the cross-section of firms

Effect of crypto surprises on crypto-exposed firms' equity prices

Average effect



NOTE. Average effect of a crypto surprise (scaled to lead to a 1% increase in Bitcoin prices at daily frequency) on firm-level equity prices. Shaded areas show 68% and 90% confidence intervals based on robust standard errors clustered two-way (at the sector-week level).

Panel local projections: Heterogeneity

- Estimate the effect of crypto surprises on equity prices across sectors

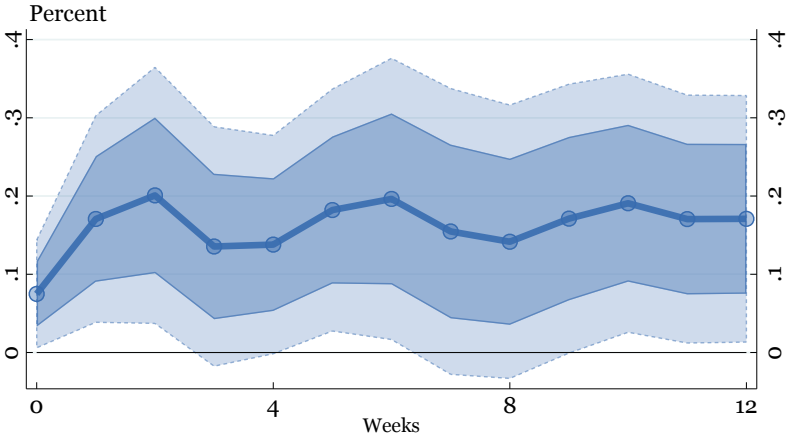
$$y_{it} = \alpha_i + \alpha_{sct,k,t} + \gamma \left(s_t \times CryptoExpo_{i,\tau-1}^{High} \right) + \sum_{p=0}^{4} \phi_p y_{i,t-1} + \varepsilon_{it}$$

  - y_{it} weekly equity price for firm i
  - α_i and $\alpha_{sct,k,t}$ are firm and time-by-sector-by-country fixed effects
  - s_t weekly crypto surprise
  - $CryptoExpo_{i\tau}^{High} = 1$ if $CryptoExpo_{i\tau}$ is above the median
  - Standard errors clustered two-way (firm and week)

- $\boxed{\gamma}$ captures the relative effect on the equity price of highly exposed firms

Effect of crypto surprises on crypto-exposed firms' equity prices

Relative effect



NOTE. Relative effect of a crypto surprise (scaled to lead to a 1% increase in Bitcoin prices at daily frequency) on firm-level equity prices for firms with high crypto exposure. Shaded areas show 68% and 90% confidence intervals based on robust standard errors clustered two-way (at the sector-week level).

Robustness

- Surprises based on 1-hour window Go
- Drop 10 largest surprises Go
- LP-IV specification Go
- Keep events with macro news overlap Go
- Time-by-sector fixed effects (relative effect only) Go
- Post-2020 sample Go

Mechanisms

Inspecting the mechanism

- **Key Questions**
  - What exactly does our crypto exposure measure pick up?
  - Who are the most exposed firms, and why?

Inspecting the mechanism

- **Key Questions**
  - What exactly does our crypto exposure measure pick up?
  - Who are the most exposed firms, and why?

- **Empirical strategy**
  - Zoom in on the top 200 most exposed firms
  - Narratively sort them by type of exposure (g) and likely transmission channel
  - Estimate group-specific sensitivity to crypto surprises:

$$y_{it} = \alpha_{i,g} + \beta_g s_t + \varepsilon_{it} \quad \text{for } g = 1, \ldots, G$$

  - Assess whether the estimated β_g differ systematically across exposure types

Types of exposure and transmission mechanisms

- Assign each of top-200 firm to one of the following groups
  - If firms span multiple groups, use the dominant one
  - Handful of firms unclassified ($\approx 10\%$)

Crypto Holders
Balance sheet effects
[e.g. MicroStrategy, Tesla]

Crypto Miners
Operating leverage
[e.g. Marathon Digital, Hut8]

Crypto Financial Services
Volume sensitivity
[e.g. Coinbase, Robinhood]

Crypto Non-Fin. Services
Tech & Adoption effects
[e.g. Metaworks]

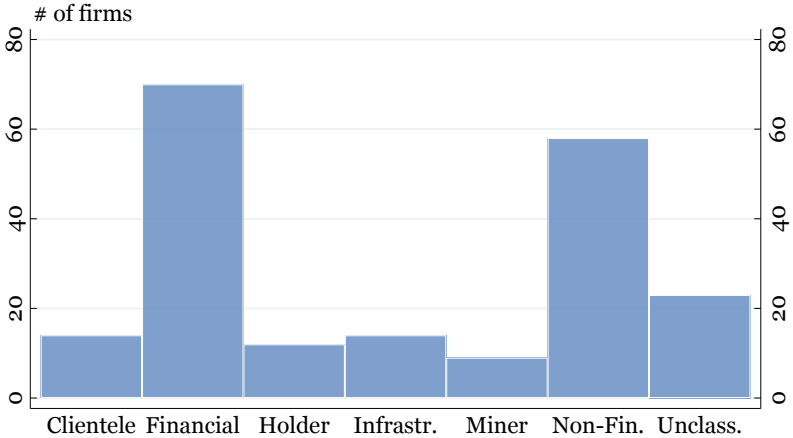
Crypto Clientele
Wealth & Sentiment effects
[e.g. Gamestop]

Crypto Infrastructure
Hardware / Cloud demand
[e.g. NVIDIA]

More details

Number of firms by type of exposure

Top-200 firms by crypto exposure



NOTE. Number of firms by type of exposure among the 200 most crypto-exposed firms. Sample period: 2017Q1–2022Q4.

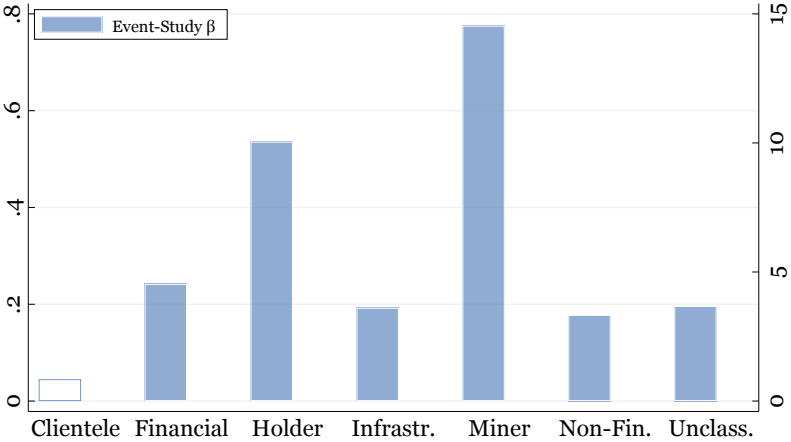
By traditional sectors

Effect of crypto surprises by type of exposure

	(1) Clientele	(2) Financial	(3) Holder	(4) Infrastruct.	(5) Miner	(6) Non Financial	(7) Unclassified
Crypto surprise (β_g)	0.05	0.24***	0.54***	0.19**	0.77***	0.18***	0.20**
	(0.10)	(0.09)	(0.12)	(0.10)	(0.21)	(0.06)	(0.08)
Observations	1,166	5,779	1,112	1,185	619	4,493	1,931
R squared	0.01	0.03	0.07	0.01	0.06	0.02	0.01
# of Firms	14	70	12	14	9	58	23
Firm FE	yes	yes	yes	yes	yes	yes	yes
Time x Sector FE	no	no	no	no	no	no	no
Time x Sector x Country FE	no	no	no	no	no	no	no

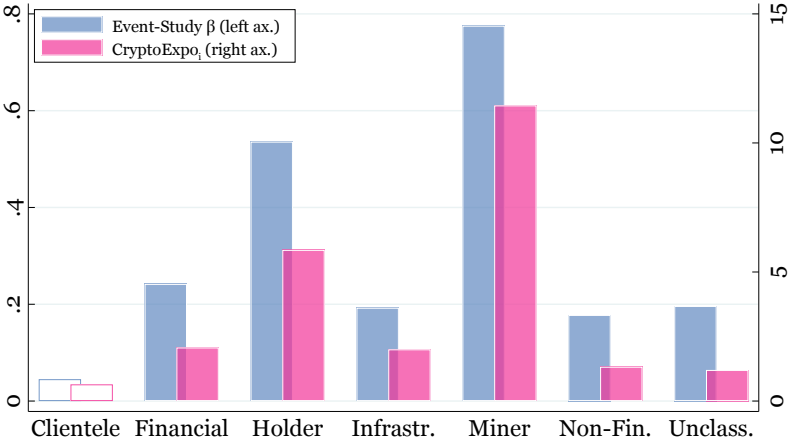
NOTE. Robust standard errors (clustered two-way, by firm and day) are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Coefficients corresponding to the constant and fixed effects not reported.

Heterogeneous sensitivity...



NOTE. Estimated response to crypto surprises on equity prices of top-200 firms grouped by type of exposure vs. cross-sectional average of the crypto-currency exposure measure $CryptoExpo_i$. Sample period: 2017q1 to 2022q4.

Heterogeneous sensitivity... or heterogeneous exposure?



NOTE. Estimated response to crypto surprises on equity prices of top-200 firms grouped by type of exposure vs. cross-sectional average of the crypto-currency exposure measure $CryptoExpo_i$. Sample period: 2017q1 to 2022q4.

Controlling for exposure levels

- Differences in sensitivity may reflect different exposure levels, not transmission mechanisms

Controlling for exposure levels

- Differences in sensitivity may reflect different exposure levels, not transmission mechanisms
- **Solution** Restrict sample to minimize heterogeneity in crypto exposure across groups
  - Grid search over $\text{CryptoExpo}_i \in [0.25, 2]$ Distribution
  - Select range that minimizes mean squared deviation from group mean
  - Optimal range is $\text{CryptoExpo}_i \in [0.5, 1.5]$

Controlling for exposure levels

- Differences in sensitivity may reflect different exposure levels, not transmission mechanisms
- **Solution** Restrict sample to minimize heterogeneity in crypto exposure across groups
  - Grid search over $\text{CryptoExpo}_i \in [0.25, 2]$ Distribution
  - Select range that minimizes mean squared deviation from group mean
  - Optimal range is $\text{CryptoExpo}_i \in [0.5, 1.5]$
- Then
  - Re-estimate regressions with now (roughly) constant exposure across groups
  - Attribute differences in β_g to mechanisms, not exposure

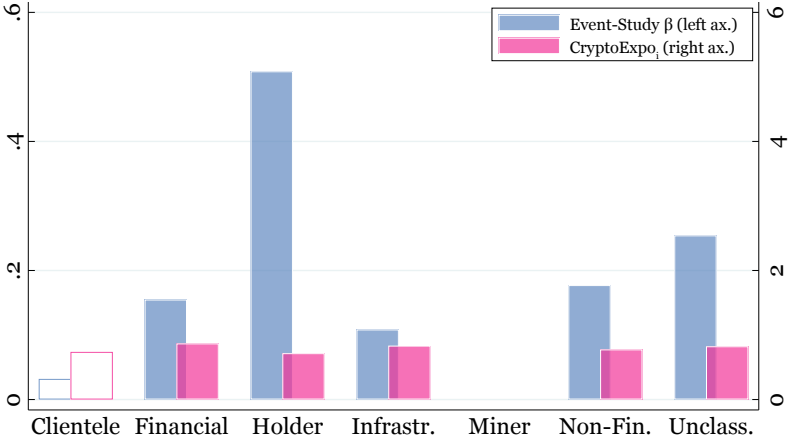
Effect of crypto surprises by type of exposure

Sample restricted to groups with similar exposure → CryptoExpo$_i \in [0.5, 1.5]$

	(1) Clientele	(2) Financial	(3) Holder	(4) Infrastruct.	(5) Miner	(6) Non Financial	(7) Unclassified
Crypto surprise (β)	0.03	0.16*	0.51**	0.11	(.)	0.18***	0.25*
	(0.20)	(0.08)	(0.20)	(0.09)	(.)	(0.06)	(0.13)
Observations	437	2,808	200	715	0	1,734	542
R squared	0.01	0.03	0.05	0.01	(.)	0.01	0.02
# of Firms	5	32	2	7	0	21	7
Firm FE	yes	yes	yes	yes	yes	yes	yes
Time x Sector FE	no	no	no	no	no	no	no
Time x Sector x Country FE	no	no	no	no	no	no	no

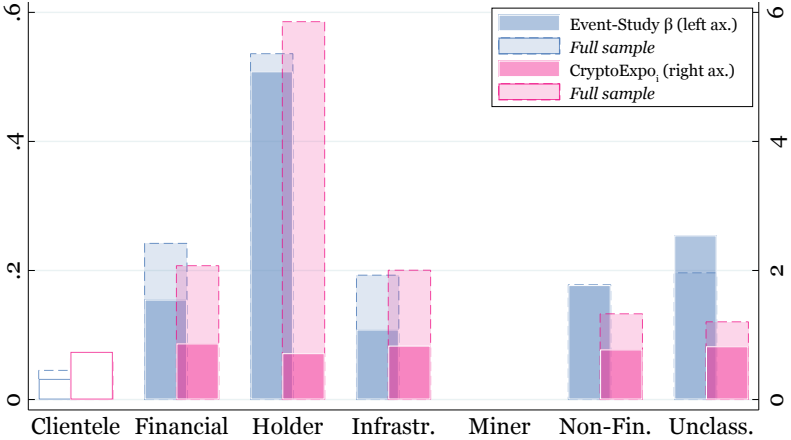
NOTE. Robust standard errors (clustered two-way, by firm and day) are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Coefficients corresponding to the constant and fixed effects not reported.

Exposure (β^g) vs. Sensitivity (CryptoExpo$_g$)



NOTE. Estimated response to crypto surprises on equity prices of top-200 firms grouped by type of exposure vs. cross-sectional average of the crypto-currency exposure measure *CryptoExpo*$_i$, based on a restricted sample of firms where cross-group exposure levels are comparable. Sample period: 2017q1 to 2022q4.

Exposure (β^g) vs. Sensitivity (CryptoExpo$_g$)



NOTE. Estimated response to crypto surprises on equity prices of top-200 firms grouped by type of exposure vs. cross-sectional average of the crypto-currency exposure measure *CryptoExpo*$_i$, based on a restricted sample of firms where cross-group exposure levels are comparable. Sample period: 2017q1 to 2022q4.

Conclusions

Concluding remarks

- Crypto is not (yet?) a pressing risk for policymakers
  - Firm, household, and traditional financial sector exposure remains limited
- But evidence on effect of crypto shocks highlights potential risks that should not be ignored
  - Monitoring the evolution of crypto exposure is key
  - Deepen understanding of crypto surprises transmission mechanism

Concluding remarks

- Crypto is not (yet?) a pressing risk for policymakers
  - Firm, household, and traditional financial sector exposure remains limited
- But evidence on effect of crypto shocks highlights potential risks that should not be ignored
  - Monitoring the evolution of crypto exposure is key
  - Deepen understanding of crypto surprises transmission mechanism
- **Next steps**
  - Investigate real firm-level outcomes beyond equity prices

Crypto Shocks

Ambrogio Cesa-Bianchi
Bank of England, CEPR, and CfM

Marco Pinchetti
Banque de France

ESSIM
29 May 2025

*The views expressed in this paper are those of the author(s) and do not necessarily represent the views of the Bank of England.

References I

Adrian, T., T. Iyer, and M. S. Qureshi (2022): "Crypto prices move more in sync with stocks, posing new risks," *IMF Blog*.

Ahir, H., N. Bloom, and D. Furceri (2022): "The World Uncertainty Index," *NBER Working Papers, 29763*.

Aiello, D., S. R. Baker, T. Balyuk, M. DiMaggio, M. J. Johnson, and J. D. Kotter (2023): "The Effects of Cryptocurrency Wealth on Household Consumption and Investment," *NBER Working Papers, 31445*.

Benigno, G. and C. Rosa (2023): "The Bitcoin–Macro Disconnect," *FRB of New York Staff Report*.

Caldara, D. and M. Iacoviello (2022): "Measuring Geopolitical Risk," *American Economic Review*, 112, 1194–1225.

Che, M. N. X., A. Copestake, D. Furceri, and T. Terracciano (2023): "The Crypto Cycle and US Monetary Policy," *IMF Working Papers*.

Cloyne, J. (2013): "Discretionary Tax Changes and the Macroeconomy: New Narrative Evidence from the United Kingdom," *American Economic Review*, 103, 1507–1528.

Cloyne, J. and P. Hurtgen (2016): "The Macroeconomic Effects of Monetary Policy: A New Measure for the United Kingdom," *American Economic Journal: Macroeconomics*, 8, 75–102.

Divakaruni, A. and P. Zimmerman (2024): "Uncovering Retail Trading in Bitcoin: The Impact of COVID-19 Stimulus Checks," *Management Science*, 70, 2066–2085.

Fernald, J. G. (2012): "A quarterly, utilization-adjusted series on total factor productivity," *Federal Reserve Bank of San Francisco Working Paper Series, 2012-19*.

Gertler, M. and P. Karadi (2015): "Monetary Policy Surprises, Credit Costs, and Economic Activity," *American Economic Journal: Macroeconomics*, 7, 44–76.

Gurkaynak, R. S., B. Kisacikoglu, and J. H. Wright (2020): "Missing Events in Event Studies: Identifying the Effects of Partially Measured News Surprises," *American Economic Review*, 110, 3871–3912.

Gürkaynak, R. S., B. Sack, and E. Swanson (2005): "Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements," *International Journal of Central Banking*, 1.

References II

Hassan, T. A., S. Hollander, L. van Lent, and A. Tahoun (2019): "Firm-Level Political Risk: Measurement and Effects," *The Quarterly Journal of Economics*, 134, 2135–2202.

Karau, S. (2023): "Monetary policy and Bitcoin," *Journal of International Money and Finance*, 137.

Känzig, D. R. (2021): "The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements," *American Economic Review*, 111, 1092–1125.

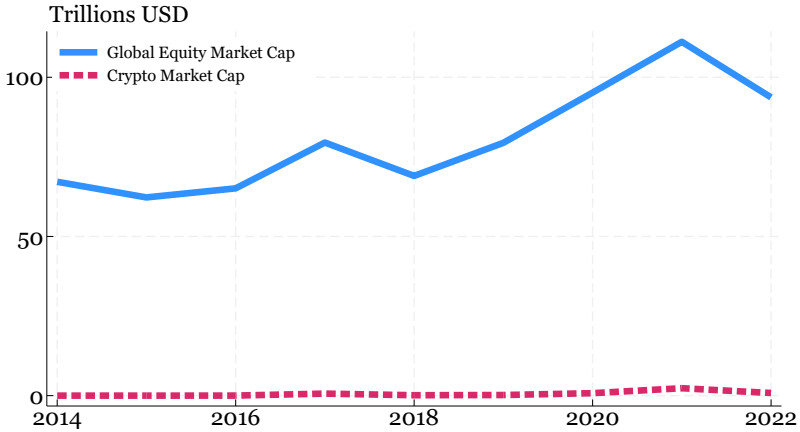
Romer, C. D. and D. H. Romer (2004): "A New Measure of Monetary Shocks: Derivation and Implications," *American Economic Review*, 94, 1055–1084.

——— (2010): "The Macroeconomic Effects of Tax Changes: Estimates Based on a New Measure of Fiscal Shocks," *American Economic Review*, 100, 763–801.

Weber, M., B. Candia, O. Coibion, and Y. Gorodnichenko (2023): "Do You Even Crypto, Bro? Cryptocurrencies in Household Finance," *NBER Working Paper Series, 31284*.

A0: Crypto Facts

Global Equity vs. Crypto Market Cap



Back

A1: Crypto Surprises

Crypto n-grams

- **Generic:** crypto, cryptos, cryptocurrency, crypto-currency, cryptocurrencies, crypto-currencies, cryptoasset, crypto-asset, cryptoassets, crypto-assets, cryptospace, crypto-space, crypto-mining, cryptomining, crypto-economy, cryptoeconomy, crypto-mining, cryptomining, cryptowallet, crypto-wallet, initial coin offering, initial coin offerings
- **Technology/Applications:** blockchain, block-chain, block chain, blockchains, block-chains, block chains, distributed ledger, DLT, DLTs, DeFi, smart contracts, NFT, NFTs, tokenization
- **Tokens:** Bitcoin, Bitcoins, BTC, BTCs, Ethereum, Ethereums, ETH, ETHs
- **Stablecoins:** stablecoin, stablecoins, USD Coin, USDC, Tether, USDT

Back

Crypto surprises: Top 10 Largest Events

Date	Title	ΔBTC
01/12/2017	CME gets regulatory approval to trade Bitcoin future contracts	+5.70%
05/09/2018	Goldman Sachs scraps project to open a crypto trading desk	-5.67%
25/04/2019	NY Attorney General announces court order against exchange Bitfinex	-7.08%
13/05/2019	Launch of network Flexa for crypto payments in mainstream stores	+6.55%
10/07/2019	Fed Governor Powell raises "serious concerns" over Facebook's Libra	-5.12%
29/01/2021	Elon Musk publicly endorses via his Twitter bio	+12.63%
08/02/2021	Tesla announces 1.5 Billion USD Bitcoin purchase	+10.78%
12/05/2021	Tesla stops accepting Bitcoin as means of payment	-11.76%
14/11/2022	Binance announces creation of "crypto recovery fund"	+5.09%

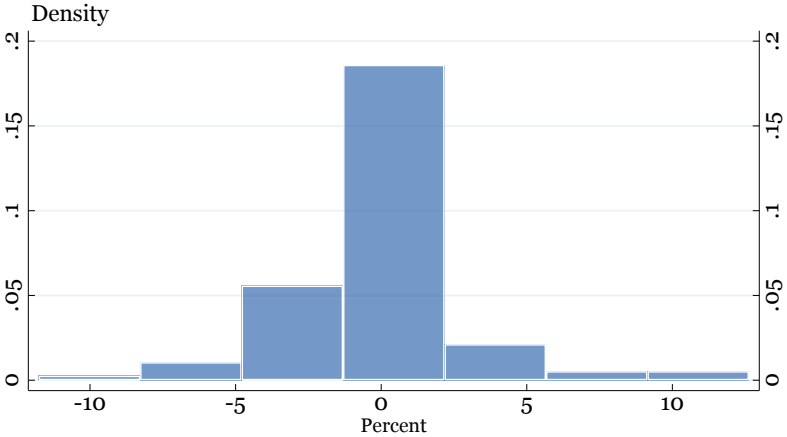
Crypto surprises: Summary Statistics

	Shock (1 hour)	Shock (2 hour)	Shock (4 hour)
Mean	0.01	-0.01	-0.02
Median	0.00	0.00	0.00
Standard Dev.	0.47	0.77	0.80
Minimum	-3.42	-11.76	-9.32
Maximum	8.32	12.63	10.07
Observations	1539	1539	1539

NOTE. Summary statistics for the daily crypto surprises. The sample covers the period between Jan 2017 and 31 Dec 2022.

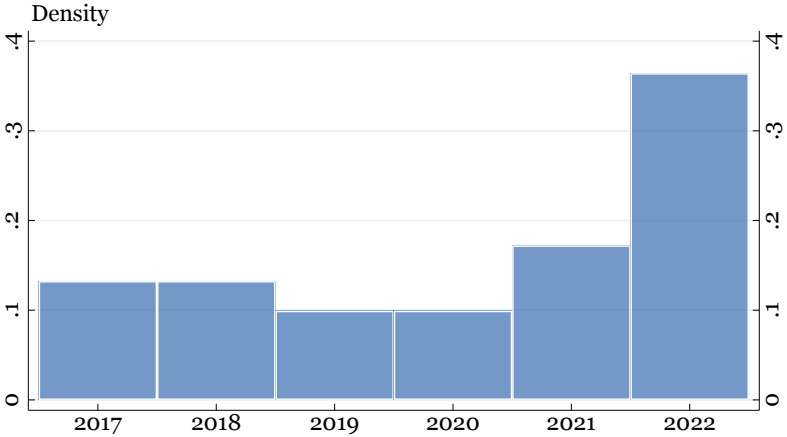
Back

Crypto surprises: Distribution



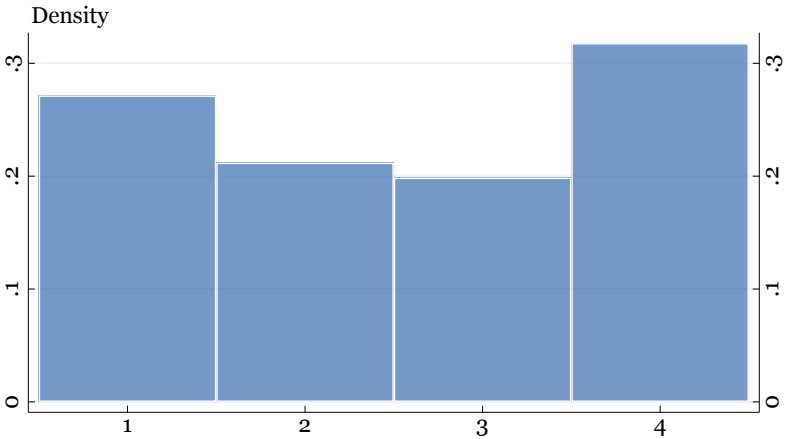
Back

Distribution of crypto surprises: By year



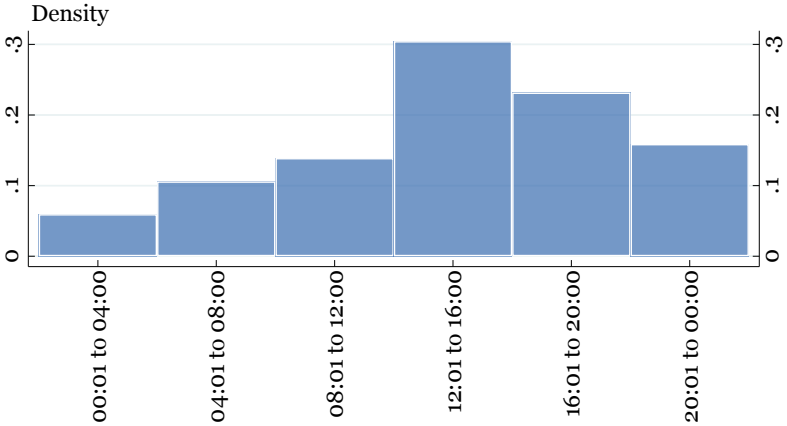
Back

Distribution of crypto surprises: By quarter



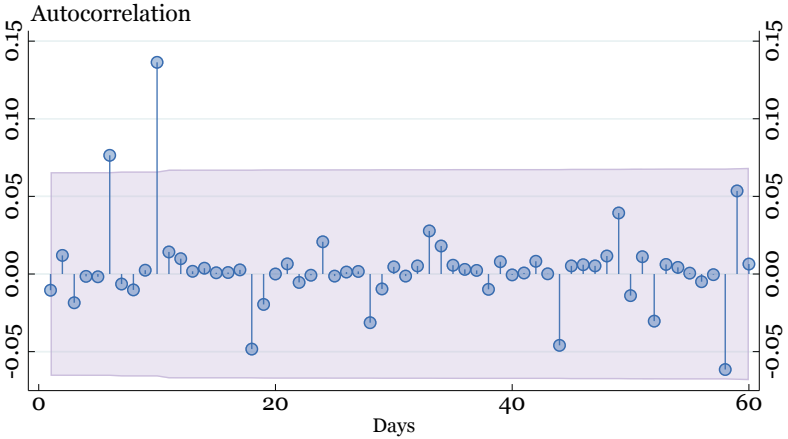
Back

Distribution of crypto surprises: By hour



Back

Crypto surprises: Autocorrelation



Back

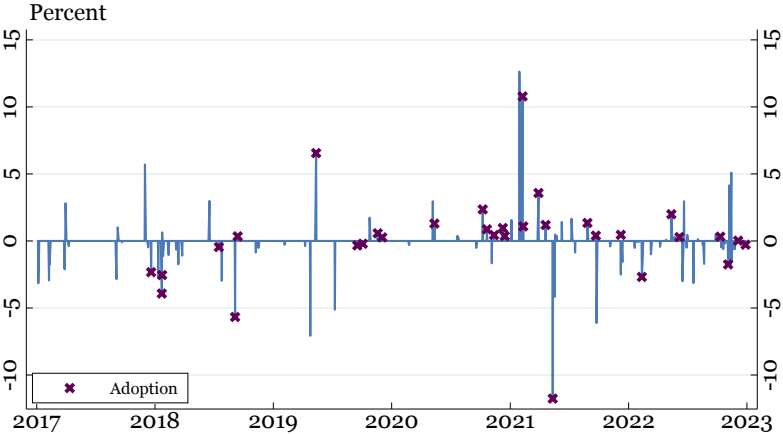
Classification of crypto surprises

- **Adoption** Company announcements signalling adoption of crypto into mainstream finance
  - Banks offering crypto custody services
  - Public companies adding Bitcoin to their balance sheets
  - Payment platforms enabling crypto transactions
- **Regulation** Policy news about the legal/institutional framework of crypto
  - SEC enforcement actions against crypto exchanges
  - New crypto taxation laws or mining bans
- **Confidence** Changes in investor perceptions or confidence in the future of cryptocurrencies
  - Major exchange hacks or protocol failures
  - Positive/negative statements by influential figures (e.g., Elon Musk tweets)

Back

The crypto surprise series

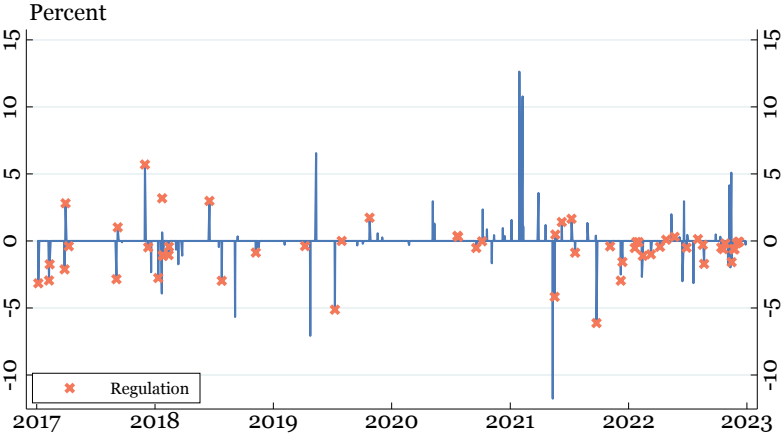
- Surprises classification → Adoption



Definitions | Stats | Distribution | By Hour | By Quarter | By Year | Autocorrelation

The crypto surprise series

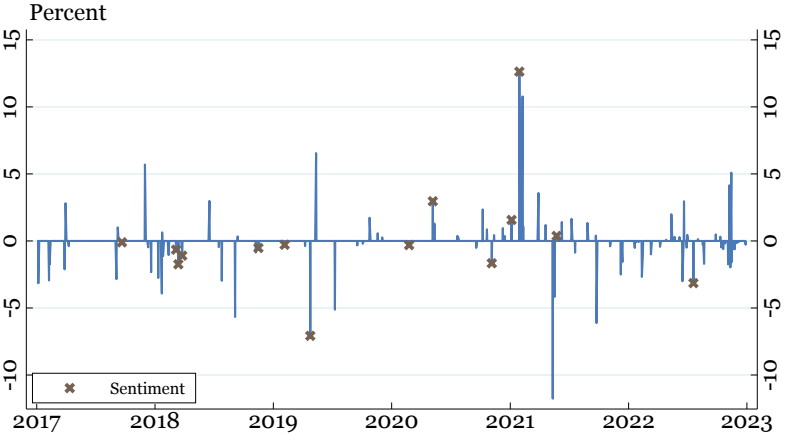
- Surprises classification → Regulation



Definitions | Stats | Distribution | By Hour | By Quarter | By Year | Autocorrelation

The crypto surprise series

- Surprises classification → Confidence



Definitions | Stats | Distribution | By Hour | By Quarter | By Year | Autocorrelation

Correlation with other shocks

Shock / Index	Source	ρ
Oil	Känzig (2021)	0.21
Monetary	Gertler and Karadi (2015)	-0.12
Geopolitical Risk	Caldara and Iacoviello (2022)	-0.14
TFP	Fernald (2012)	0.07
World Uncertainty	Ahir, Bloom, and Furceri (2022)	-0.08

A2: Crypto Exposure

NL Analytics

- **What is it**: automated text-based earnings calls search tool
- **Academic foundations**: Hassan et al. (2019) and subsequent papers
- **Dataset**: earnings calls' transcripts from listed firms worldwide (LSEG-Refinitiv) from 2002
- **Coverage (% of mkt cap)**: 96% in the US, 90% in the EA, 90% in the UK, 35% in China
- **Location**: Identified by firms' headquarters
- **Input**: Set of thematic keywords (e.g. supply chains, Brexit, etc)
- **Output**: quarterly firm-level measure based on a query defined by input keywords

Back

Measuring crypto risk

- For a given earning call c, crypto risk is the ratio of sentences mentioning crypto n-grams which contain expressions associated with risk over the total number of sentences:

$$\text{CryptoRisk}_{cit} = \frac{1}{N_{cit}} \sum_{n=1}^{N_{cit}} \{\mathbb{1}\left[n_{cit} \in \mathbb{C}\right] \times \mathbb{1}\left[n_{cit} \in \mathbb{R}\right]\}$$

where $n = 0, 1, \ldots, N_{cit}$ are the words contained in the call c of firm i in quarter t, and $\mathbb{R}$ is the set of synonyms for risk, risky, uncertain, or uncertainty

Back

Measuring crypto sentiment

- For a given earning call c, crypto sentiment is the ratio of the difference between crypto sentences with a positive tone minus n-grams with a negative tone over total sentences:

$$\text{CryptoSent}_{cit} = \frac{1}{N_{cit}} \sum_{n=1}^{N_{cit}} \{\mathbb{1}\,[n_{cit} \in \mathbb{C}] \times \text{Tone(n)}\}$$

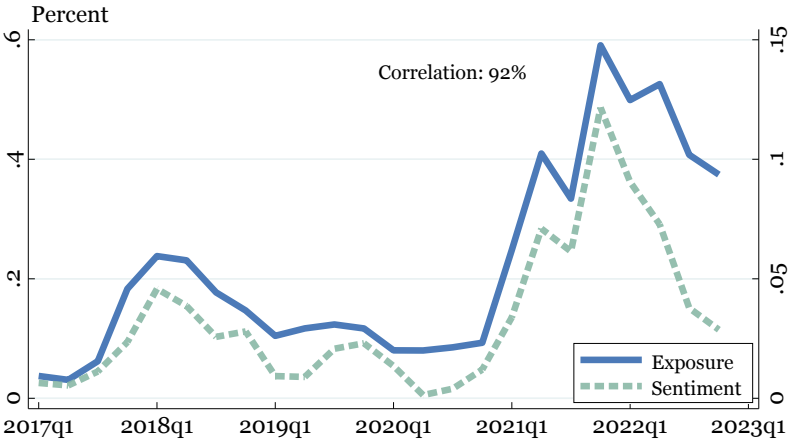
where Tone(s) maps a sentence triple's sentiment words into integer counts, by returning:

  - The sum of positive words if the sentence triple contains positive but not negative words
  - The negative sum of negative words if the sentence triple contains negative but not positive words
  - Zero if the sentence triple contains both positive and negative words

Back

Crypto exposure vs. sentiment measure

Average across firms

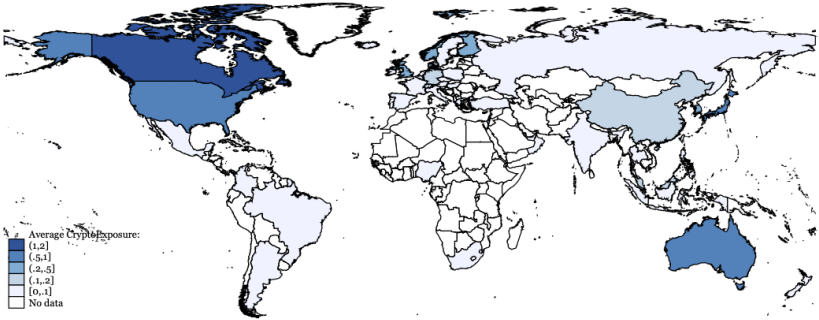


NOTE. Cross-sectional average (with equal weights) of the crypto-currency exposure measure $CryptoExpo_{it}$ and sentiment measure $CryptoSent_{it}$ across all firms in our sample. Sample period: 2017q1 to 2022q4.

Back

Crypto exposure measure

Geographic breakdown: Exposure



Back

Overlap Between Crypto and US Equity Markets

Market Cap

Year	Crypto MktCap / S&P500 MktCap	Crypto MktCap / Total US MktCap
2017	34.05%	25.65%
2018	35.17%	25.49%
2019	36.88%	30.60%
2020	41.75%	34.22%
2021	41.96%	34.01%
2022	33.87%	28.24%

Back

Overlap Between Crypto and US Equity Markets

Number of Firms

Year	# Crypto Firms / # S&P500 Firms	# Crypto Firms / # Total US Market
2017	23.09%	24.80%
2018	26.14%	25.00%
2019	27.70%	25.20%
2020	27.27%	25.20%
2021	24.59%	25.20%
2022	25.05%	25.40%

Back

A3: Event Study

The effect of crypto surprises on firm-level equity prices

Linear interaction

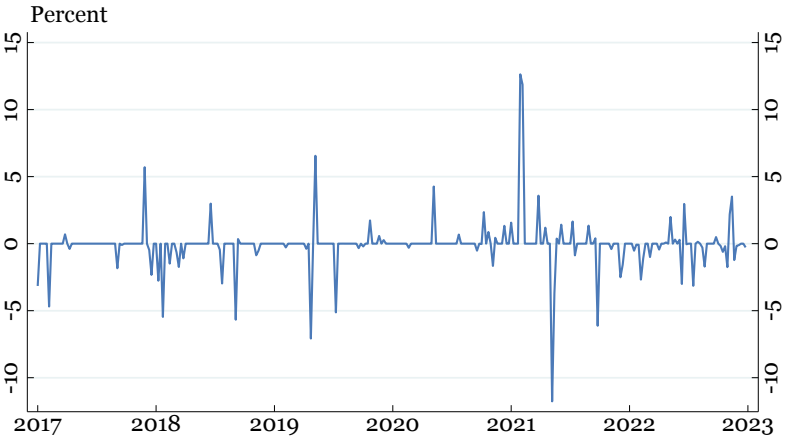
	(1)	(2)	(3)	(4)
Crypto surprise (β)	0.10**	0.11**		
	(0.05)	(0.05)		
Crypto surprise x Exposure (γ)		0.04*	0.03*	0.03*
		(0.02)	(0.02)	(0.02)
Exposure (δ)		-0.13**	-0.11***	-0.12***
		(0.06)	(0.04)	(0.04)
Observations	113,006	108,085	108,085	98,458
R squared	0.02	0.02	0.20	0.32
Firm FE	yes	yes	yes	yes
Time x Sector FE	no	no	yes	no
Time x Sector x Country FE	no	no	no	yes

NOTE. Robust standard errors (clustered two-way, by firm and day) are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Coefficients corresponding to the constant and fixed effects not reported.

A5: Local Projections

Weekly crypto surprises

- Collapse daily data set by week → 312 weeks from 2017w1 to 2022w52



Back

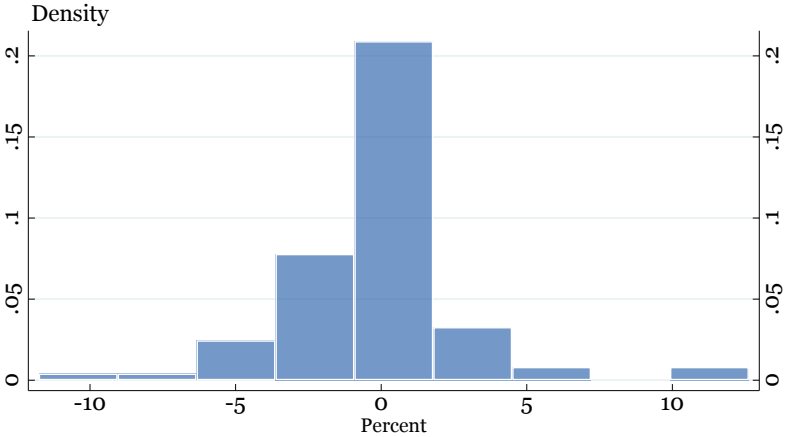
Weekly crypto surprises: Summary statistics

	Shock (1 hour)	Shock (2 hour)	Shock (4 hour)
Mean	0.04	-0.07	-0.09
Median	0.00	0.00	0.00
Standard Dev.	1.02	1.71	1.74
Minimum	-4.39	-11.76	-8.87
Maximum	9.60	12.63	9.94
Observations	312	312	312

NOTE. Summary statistics for the weekly surprises (sum of daily surprises over each week). The sample covers the period between 2017w1 and 2022w52.

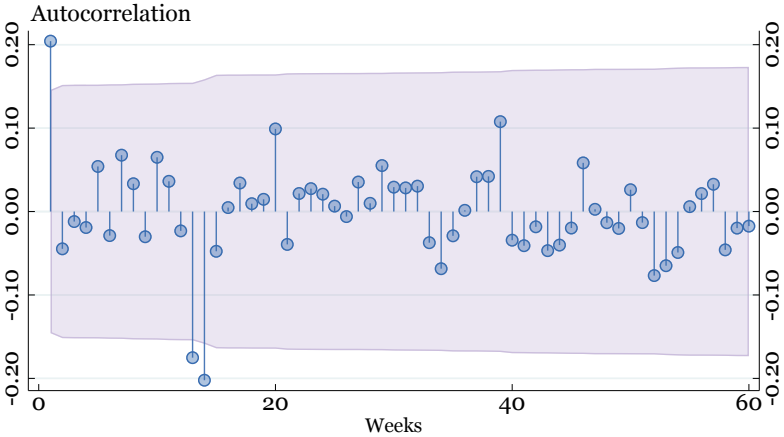
Back

Weekly crypto surprises: Distribution



Back

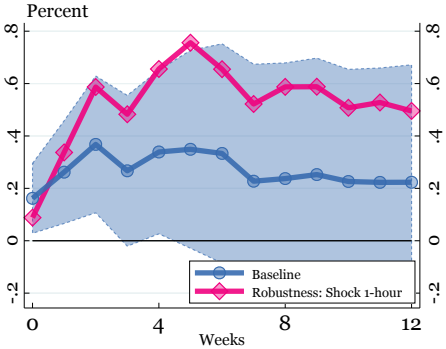
Weekly crypto surprises: Autocorrelation



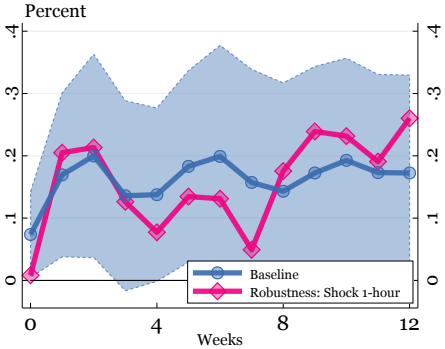
Back

Local Projections: Shock based on 1-hour window

Panel A: Average effect (β)



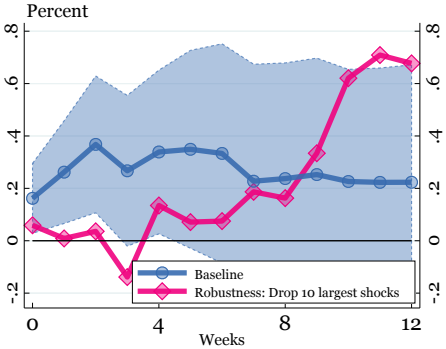
Panel B: Relative effect (γ)



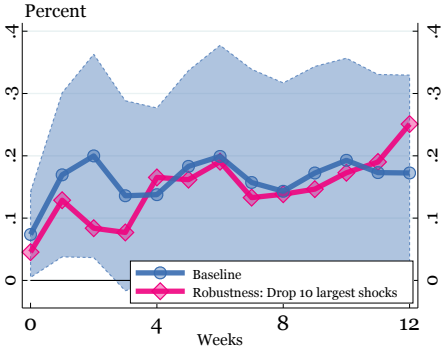
Back

Local Projections: Drop 10 biggest surprises

Panel A: Average effect (β)



Panel B: Relative effect (γ)



Back

Local Projections: IV specification

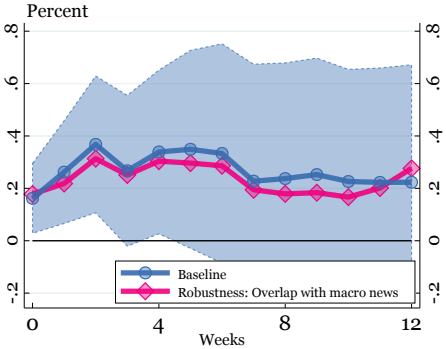
Panel A: Average effect (β)

Panel B: Relative effect (γ)

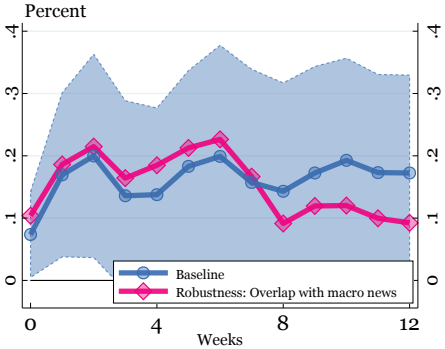
Back

Local Projections: Overlap with macro news

Panel A: Average effect (β)



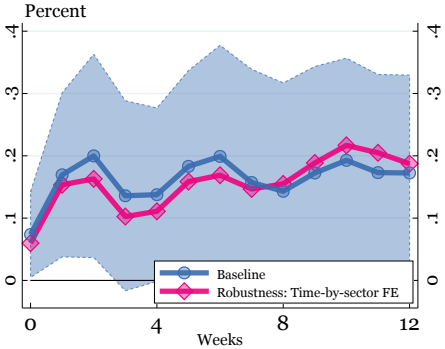
Panel B: Relative effect (γ)



Back

Local Projections: Time-by-Sector fixed effects

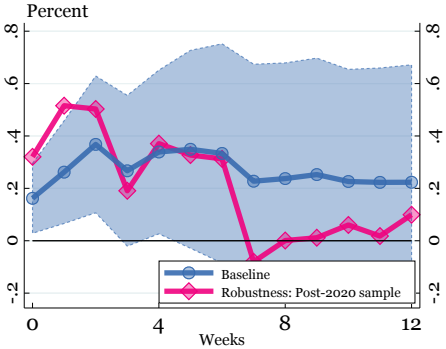
Panel B: Relative effect (γ)



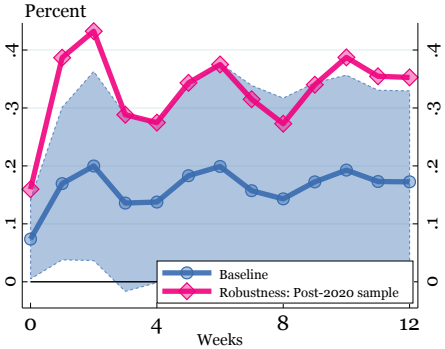
Back

Local Projections: Post-2020 sample

Panel A: Average effect (β)



Panel B: Relative effect (γ)



Back

A4: Zooming into Crypto-Firms

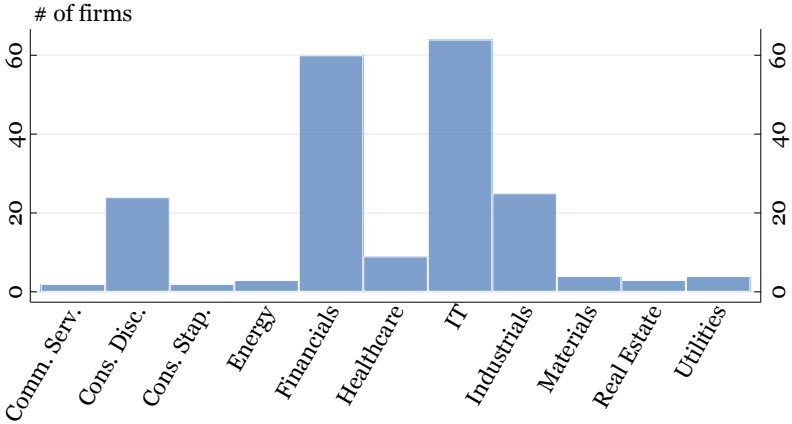
Transmission Channels by Type

- **Crypto Holders** → Balance sheet effects [MicroStrategy, Tesla,...]
  - Changes in crypto prices affect firms holding digital assets directly
- **Crypto Miners** → Operating leverage [Marathon Digital, Hut 8,...]
  - Rising profits due to partially fixed costs and variable revenues
- **Crypto Financial Services** → Transaction volume sensitivity [Coinbase, Robinhood,...]
  - Trading platforms' profits rise with higher crypto volumes
- **Crypto Non-Financial Services** → Technology & Adoption effects [Metaworks,...]
  - Technological Spillovers & Product Variety Expansion
- **Crypto Clientele** → Client wealth & sentiment [GameStop,...]
  - Retail-driven firms see higher demand linked to wealth and sentiment effects
- **Crypto Infrastructure Providers** → Infrastructure demand [NVIDIA,...]
  - Rising hardware / cloud demand for crypto-related computing power

Back

Number of firms by sector

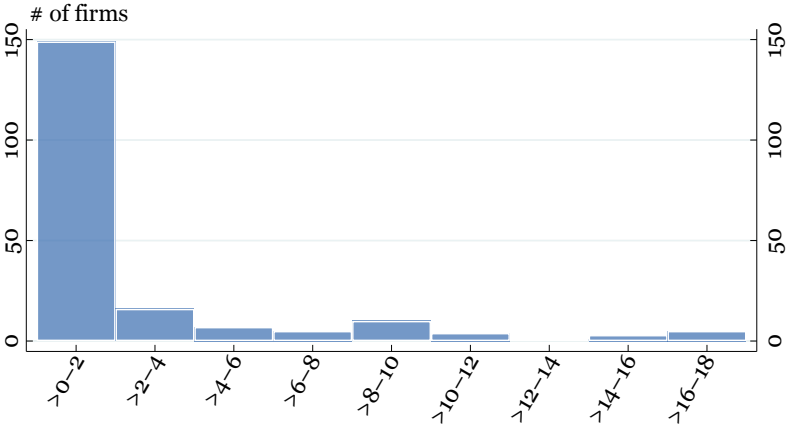
Top-200 firms by crypto exposure



NOTE. Number of firms by sector among the 200 most crypto-exposed firms. Sample period: 2017Q1–2022Q4.

Back

Distribution of crypto exposure across top-200 firms



NOTE. Binned distribution of the crypto-currency exposure measure $CryptoExpo_i$ exposure for top-200 firms. Sample period: 2017q1 to 2022q4.

Back